

ESP32 Embedded Systems Lab - Student Work Log

Patched Exercise Guide with Installation Notes & Troubleshooting

ESP32 DevKit V1 - Blink LED Exercise

Student Session: June 16, 2026

Board: DOIT ESP32 DevKit V1 (ESP32-D0WD-V3)

SECTION 1: Arduino IDE Environment Setup

Step 1: Install ESP32 Board Package

Open Arduino IDE	Version 2.3.10 used
File -> Preferences	Open preferences dialog
Additional Board Manager URL	Paste this exact URL:
https://espressif.github.io/arduino-esp32/package_esp32_index.json	
Click OK	Save preferences
Tools -> Board -> Boards Manager	Open board manager
Search 'ESP32'	Install 'ESP32 by Espressif Systems'
Wait for download	~200MB download, may take several minutes

Step 2: Install USB-to-UART Driver (CP2102)

Download CP210x driver from Silicon Labs:
<https://www.silabs.com/developers/usb-to-uart-bridge-vcp-drivers>
Run CP210xVCPInstaller_x64.exe (Windows)
Restart computer after installation
Verify in Device Manager: 'Silicon Labs CP210x USB to UART Bridge (COMx)'

Step 3: Select Correct Board & Port

Tools -> Board	ESP32 Arduino -> 'DOIT ESP32 DEVKIT V1'
Tools -> Port	Select COM port (e.g., COM3)
Tools -> Upload Speed	921600 (default)
Verify compilation	Click checkmark button
Upload sketch	Click arrow button

Upload Troubleshooting:

If upload hangs on 'Connecting...', press and hold BOOT button, then press EN button, release EN, then release BOOT. This forces bootloader mode.

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SECTION 2: Hardware Configuration & Issues

Working Hardware

Board:	DOIT ESP32 DevKit V1 - Functional
On-board LED:	GPIO 2 (blue LED) - Working perfectly
USB Cable:	Data + Power cable - Verified
Breadboard:	Half-size solderless breadboard

Hardware Issues Encountered

External LEDs (2x):	Not tested - missing 220 ohm resistors
RGB LED Module (KY-009):	DEFECTIVE - tested on GPIO 18/19/21 with GND and 3V3, no light output
Resistors available:	Only 22 ohm (red-red-black) - too low for LEDs
Push button:	Wiring tested but not functional with available components

Root Cause Analysis

The RGB LED module (KY-009) was tested exhaustively: all three color pins (R, G, B) were connected to GPIO 18, 19, and 21 respectively. The common pin (Clr) was tested with both GND and 3V3. The ESP32 code executed correctly (Serial Monitor confirmed GPIO state changes), but no physical light was emitted. Conclusion: the RGB module is defective (likely burned out from previous testing with incorrect resistor values or static discharge).

The available resistors (22 ohm) are insufficient for safe LED operation. A 220 ohm resistor (red-red-brown) is required for standard 5mm LEDs at 3.3V to limit current to ~10-20mA.

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SECTION 3: Final Exercise - Pattern Blink with On-Board LED

Exercise: Multi-Pattern LED Blink

Objective: Create a repeating LED pattern using the on-board blue LED (GPIO 2) since external hardware is unavailable. The pattern consists of: 2 quick blinks (200ms), 2 long blinks (800ms), steady ON for 2 seconds, steady OFF for 2 seconds.

Wiring

No external wiring required. The on-board LED is pre-connected to GPIO 2. Simply connect the ESP32 to the computer via USB cable.

Complete Source Code

```
/*
 * ESP32 Embedded Systems Lab - Exercise (Patched)
 * Student Work - June 16, 2026
 *
 * Hardware: DOIT ESP32 DevKit V1
 * LED: On-board blue LED (GPIO 2) - external RGB module defective
 *
 * Pattern: 2 quick blinks -> 2 long blinks -> on 2s -> off 2s -> repeat
 */

const int ledPin = 2;           // On-board LED pin (GPIO 2)
const int quickOn = 200;        // Quick blink ON duration (ms)
const int quickOff = 200;       // Quick blink OFF duration (ms)
const int longOn = 800;         // Long blink ON duration (ms)
const int longOff = 800;        // Long blink OFF duration (ms)
const int holdOn = 2000;        // Steady ON duration (ms)
const int holdOff = 2000;       // Steady OFF duration (ms)

unsigned long cycleCount = 0;   // Track completed cycles

void setup() {
  pinMode(ledPin, OUTPUT);
  Serial.begin(115200);

  // Startup banner
  Serial.println("=====");
  Serial.println("ESP32 LED Pattern - Student Lab Work");
  Serial.println("Board: DOIT ESP32 DevKit V1");
  Serial.println("LED: On-board GPIO 2 (external RGB dead)");
  Serial.println("Pattern: 2 quick -> 2 long -> 2s on -> 2s off");
  Serial.println("=====");
  Serial.println();
}

void blinkQuick() {
  digitalWrite(ledPin, HIGH);
  Serial.println("[QUICK] LED ON (200ms)");
  delay(quickOn);
  digitalWrite(ledPin, LOW);
  Serial.println("[QUICK] LED OFF (200ms)");
  delay(quickOff);
}
```

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```
void blinkLong() {
    digitalWrite(ledPin, HIGH);
    Serial.println("[LONG] LED ON (800ms)");
    delay(longOn);
    digitalWrite(ledPin, LOW);
    Serial.println("[LONG] LED OFF (800ms)");
    delay(longOff);
}

void holdSteady() {
    digitalWrite(ledPin, HIGH);
    Serial.println("[HOLD] LED ON (2000ms)");
    delay(holdOn);
    digitalWrite(ledPin, LOW);
    Serial.println("[HOLD] LED OFF (2000ms)");
    delay(holdOff);
}

void loop() {
    cycleCount++;
    Serial.print(">>> CYCLE #");
    Serial.println(cycleCount);
    Serial.println("-----");

    // 2 quick blinks
    blinkQuick();
    blinkQuick();

    // 2 long blinks
    blinkLong();
    blinkLong();

    // Steady on 2s, off 2s
    holdSteady();

    Serial.println("-----");
    Serial.println();
}
```

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SECTION 4: Verified Serial Monitor Output

The following output was captured from the Arduino IDE Serial Monitor at 115200 baud, confirming successful execution:

```
=====
ESP32 LED Pattern - Student Lab Work
Board: D0IT ESP32 DevKit V1
LED: On-board GPIO 2 (external RGB dead)
Pattern: 2 quick -> 2 long -> 2s on -> 2s off
=====

>>> CYCLE #1
-----
[QUICK] LED ON (200ms)
[QUICK] LED OFF (200ms)
[QUICK] LED ON (200ms)
[QUICK] LED OFF (200ms)
[LONG] LED ON (800ms)
[LONG] LED OFF (800ms)
[LONG] LED ON (800ms)
[LONG] LED OFF (800ms)
[HOLD] LED ON (2000ms)
[HOLD] LED OFF (2000ms)
-----

>>> CYCLE #2
-----
[QUICK] LED ON (200ms)
[QUICK] LED OFF (200ms)
... (pattern repeats indefinitely)
```

SECTION 5: Lessons Learned & Troubleshooting Notes

Key Takeaways

1. Always verify resistor values with a multimeter or color code chart before use.
2. Red-Red-Black = 22 ohm (TOO LOW for LEDs). Red-Red-Brown = 220 ohm (CORRECT).
3. The ESP32 on-board LED (GPIO 2) is invaluable for verifying code without external hardware.
4. Serial Monitor (115200 baud) is essential for debugging - always include Serial output in sketches.
5. When external hardware fails, isolate variables: test GPIO pins with known-good components first.
6. Unplug USB before making wiring changes to prevent short circuits.
7. KY-009 RGB modules with built-in resistors (R1, R2, R3) still require correct polarity and common pin identification.

Completed Exercises

Exercise 1: Single LED Blink - Completed with on-board LED (GPIO 2)

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Exercise 3: Serial Commands - Completed with ON/OFF control via Serial Monitor

Custom Exercise: Multi-pattern blink with structured functions and cycle counting

Pending (Requires Additional Components)

Exercise 2: Two LEDs with different delays - needs 220 ohm resistors + working LEDs

Exercise 4: Push-button control - needs functional button + pull-up/pull-down configuration

Challenge A: PWM Fade - needs working external LED or verification of GPIO capability

Document generated from student session - June 16, 2026

ESP32 DevKit V1 | Arduino IDE 2.3.10 | COM3 | 115200 baud